

June 9, 2026

Dear Professor Satchell,

Please consider the attached manuscript, “Finite-Sample Limitations of VaR Backtesting under Ex-Ante Volatility Regimes,” for publication in the Journal of Risk Model Validation.

Ex-ante volatility regimes for Mexican FIBRAs leave only 62–95 testable observations. At $n = 62$, the Kupiec test rejects a correctly specified model 7.6% of the time (nominal 5%) and has 43% power against a model breaching at 10%. The Christoffersen test is uninformative. The paper proposes replacing asymptotic tests with exact binomial inference and a conservatism metric $C = \hat{p} - \alpha$, neither of which depends on sample size. Across five equity tickers, GARCH- t consistently calibrates near the nominal rate (pooled 3.6%) while XGBoost-Pure over-breaches (9.7%, $p < 0.0001$).

The manuscript has not been published elsewhere nor submitted to another journal. A separate title page with identifying information is included for the blind review process. Data and code will be released after acceptance.

The short history of many emerging-market instruments makes the finite-sample problem studied here directly relevant to risk managers and regulators working with those assets. More broadly, any validation exercise constrained by ex-ante regime definitions, new-product histories, or regulatory look-back windows faces the same structural limitation: below 100 observations, standard backtests lose discriminatory power. The decision framework proposed in Section 6 offers a practical alternative that requires no additional data and no asymptotic approximations.

Sincerely,

Germán Pancardo